

Programming a 2D Ising Model/The effect of system size

Contents

hide

Beginning

6. The effect of system size

Scaling the System Size

TASK 6:

Page

Discussion

Read

View source

View history

Tools

This is the sixth section of the Ising model experiment.

Main Page: Programming a 2D Ising Model

Previous section: 5. The effect of temperature

Next section: 7. Determining the heat capacity

6. The effect of system size

Scaling the System Size

Your plots from the previous section showed how the magnetisation and energy vary with temperature, and you should be able to see the onset of the phase transition. In this region, the energetic and entropic driving forces are of almost equal importance, and large fluctuations in the state of the system can take place. In fact, it is a characteristic of phase transitions that fluctuations within a system start to take place over very long ranges. This is a big problem in our simulation — so far, we only have 64 spins (or "atoms", if you prefer), so our system may not be big enough for these long ranged fluctuations to be correctly modelled.

TASK 6:

Repeat the final task of the previous section for the following lattice sizes: 2×2, 4×4, 8×8, 16×16, 32×32. Make sure that you name each datafile that you produce after the corresponding lattice size! Write a Python script to make a plot showing the energy *per spin* versus temperature for each of your lattice sizes. Hint: the NumPy `loadtxt` function is the reverse of the `savetxt` function, and can be used to read your previously saved files into the script. Repeat this for the magnetisation. As before, use the plot controls to save your an SVG or PNG image of your plot and attach this to the report. How big a lattice do you think is big enough to capture the long range correlations?

This is the sixth section of the Ising model experiment.

Main Page: Programming a 2D Ising Model

Previous section: 5. The effect of temperature

Next section: 7. Determining the heat capacity

Appearance

hide

Text

- ☒

Small
- ☐

Standard
- ☐

Large

Width

- ☒

Standard
- ☐

Wide